A US Treasury Curve Model and Bond Portfolio Scenario Tool

Investment Research Portfolio · Project 11 · Curve date 30 September 2026

Summary

I build the US Treasury zero curve on 30 September 2026 from the Treasury's par yields, fit a Nelson-Siegel-Svensson (NSS) model that matches the Federal Reserve's Gürkaynak-Sack-Wright (GSW) curve within 3.9 bp at 2, 5, 10 and 30 years, and track the level, slope and curvature of the curve daily since 2006. Three factors explain 97.9% of daily yield changes (level alone 84.5%). I then price a 7-year bullet, a duration-matched 2/30-year barbell and a 1–10-year ladder ($100m each) and revalue them under six instant curve scenarios. The barbell gains 268 bp more than the bullet in a bull flattener and loses 193 bp more in a bear steepener; its extra convexity adds about 34 bp in a ±100 bp parallel move. The price of that convexity is 19 bp less 12-month carry and roll-down than the bullet if the curve does not move.

1. Data

- **U.S. Treasury Daily Par Yield Curve Rates** (home.treasury.gov), January 2006 – 30 September 2026, 5,192 days. On 30 September 2026 the curve has 14 points from 1 month to 30 years (1-month 4.02%, 2-year 4.88%, 10-year 5.29%, 20-year 5.68%, 30-year 5.64%).
- **Federal Reserve GSW dataset** (feds200628.csv, Gürkaynak, Sack and Wright, 2007): fitted Svensson parameters and continuously compounded zero-coupon yields, used as an independent benchmark.
- **NBER recession indicator** (FRED USREC) for shading.

2. Bootstrapping the zero curve

Par yields are interpolated with a natural cubic spline onto a semiannual grid from 0.5 to 30 years. Treating each grid point as a par bond paying semiannual coupons, discount factors are solved sequentially: $d(T) = (1 - c/2 \cdot \Sigma_{t<T}\, d(t)) / (1 + c/2)$. Between grid points the log discount factor is interpolated linearly. The resulting curve reprices every par bond on the grid to within 2e-12 bp.

Maturity	Par yield	Discount factor	Zero (semiannual)	Zero (continuous)
0.5y	nan	nan	nan	nan
1y	nan	nan	nan	nan
2y	nan	nan	nan	nan
3y	nan	nan	nan	nan
5y	nan	nan	nan	nan
7y	nan	nan	nan	nan
10y	nan	nan	nan	nan
20y	nan	nan	nan	nan
30y	nan	nan	nan	nan

Table 1. Bootstrapped curve on 30 September 2026 (selected maturities).

The 20-year par yield (5.68%) is above the 30-year (5.64%), a long-standing feature of the 20-year sector. Because the spline must pass through every quote, the bootstrapped zero curve bulges between 15 and 25 years (Figure 1). Bootstrapping is exact but inherits every quirk of the input quotes; a parametric curve smooths them.

3. Nelson-Siegel-Svensson fit and comparison with the Fed curve

The Svensson (1994) form, $z(t) = \beta0 + \beta1 \cdot L1(t/\tau1) + \beta2 \cdot C(t/\tau1) + \beta3 \cdot C(t/\tau2)$, is fitted by nonlinear least squares so that the model's par yields match the 14 observed par yields (bills on a bond-equivalent basis), from 18 starting points with $\beta0 \geq 0$. The fit error is 2.8 bp root-mean-square.

Parameter	This fit	Fed GSW
β0	0.000%	0.000%
β1	3.960%	3.928%
β2	4.985%	5.270%
β3	16.953%	17.093%
τ1	2.05	1.86
τ2	16.16	15.70

Table 2. Fitted Svensson parameters vs. the Fed's GSW parameters for the same date.

Zero rate	NSS (this fit)	Bootstrap	Fed GSW	NSS − GSW	Bootstrap − GSW
2y	4.80%	4.83%	4.84%	-3.9 bp	-0.8 bp
5y	5.06%	5.04%	5.04%	+1.4 bp	-0.0 bp
10y	5.30%	5.26%	5.28%	+1.2 bp	-2.3 bp
30y	5.67%	5.62%	5.66%	+0.5 bp	-4.8 bp

Table 3. Continuously compounded zero rates on 30 September 2026.

The independent fit lands very close to the Fed's: β1, β3 and τ2 are within a few percent of GSW's values and β0 sits on its zero bound in both fits, a sign that β0 and β3 are only weakly identified when the curve is humped at the long end. Fitted zero rates differ from GSW by 0.5–3.9 bp. The remaining gap reflects different inputs: GSW fits off-the-run coupon securities and excludes bills, while this fit uses the Treasury's on-the-run par curve including bills.

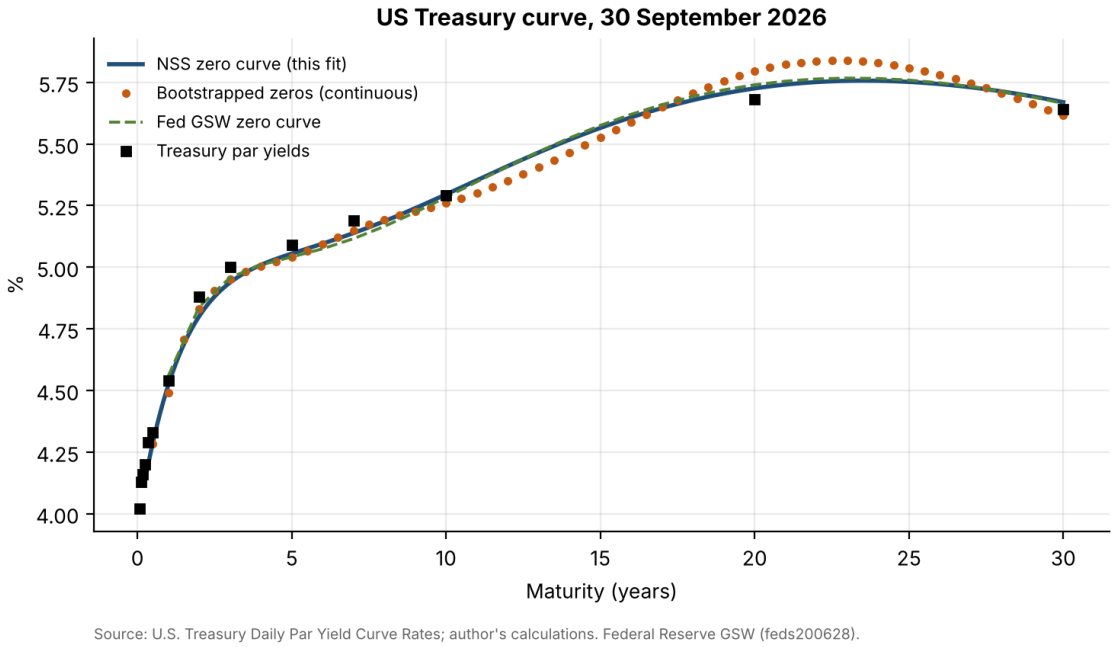
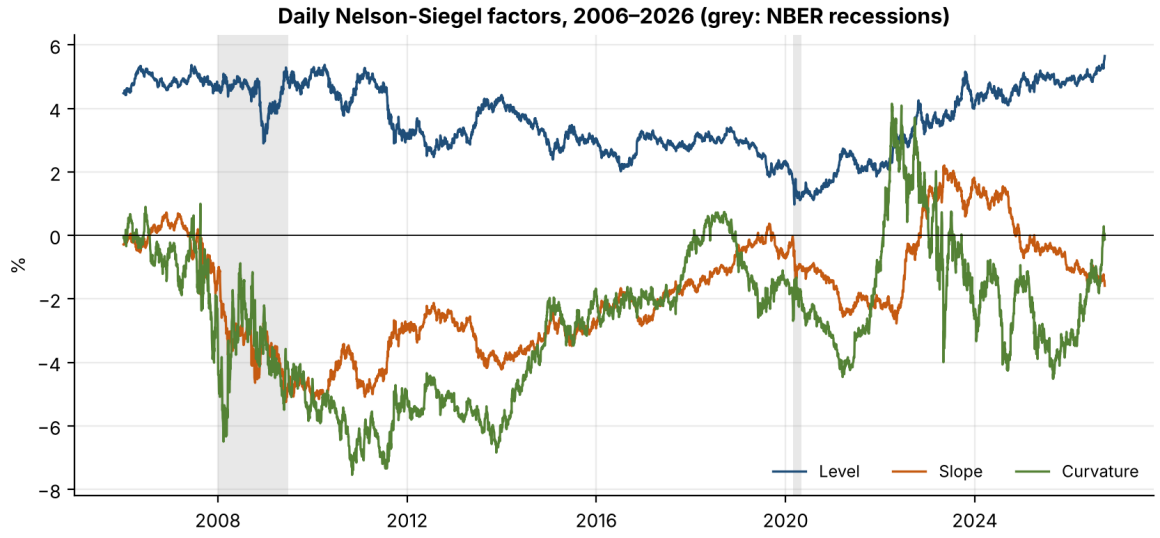


Figure 1. Par yields, bootstrapped zeros, the NSS fit and the Fed GSW curve.

4. Level, slope and curvature since 2006

Each day I fit a three-factor Nelson-Siegel model to the par yields by ordinary least squares with the decay fixed at Diebold and Li's (2006) value (0.0609 per month). Over 5,191 days the median fit error is 9.4 bp. The slope factor here is β1 (short minus long), so a negative value means an upward-sloping curve.



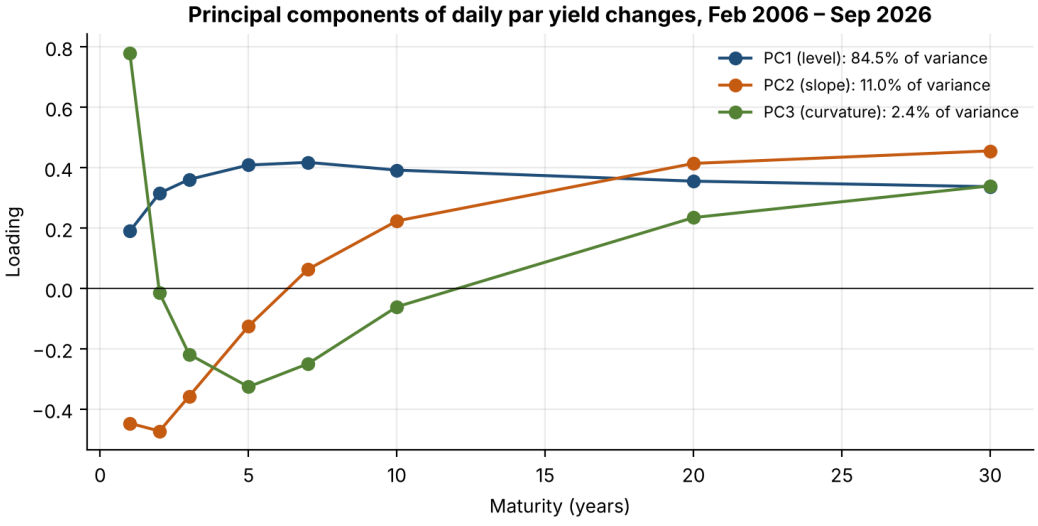
Source: U.S. Treasury Daily Par Yield Curve Rates; author's calculations. NBER via FRED USREC. Diebold-Li fixed decay (0.0609/month).

Figure 2. Daily Nelson-Siegel factors with NBER recessions shaded.

The level factor fell from about 4.9% in 2006 to a low of 1.0% in March 2020 and has risen to 5.6% by September 2026. The slope factor turned positive (an inverted curve) in October 2022 during the Fed's hiking cycle, and curvature swings most around policy turning points. Both recessions in the sample were preceded by an inverted curve (slope above zero) within the previous 12 months: up to 0.69 points in 2007 and 0.37 points in 2019, the classic inversion signal.

5. Principal components of daily yield changes

Applied to 5,163 daily changes in the 1, 2, 3, 5, 7, 10, 20 and 30-year par yields, PCA finds the three factors of Litterman and Scheinkman (1991): a level factor (84.5% of variance) with same-sign loadings at every maturity, a slope factor (11.0%) that moves short and long yields in opposite directions, and a curvature factor (2.4%) that moves the belly against the wings. Together they explain 97.9%.



Source: U.S. Treasury Daily Par Yield Curve Rates; author's calculations.

Figure 3. Loadings of the first three principal components.

6. Three $100m portfolios

Portfolio	Holdings (face, coupon)	Modified duration	Convexity	Avg. coupon
Bullet	7y 100.0m @ 5.190%	5.81	40.1	5.19%
Barbell	2y 68.6m @ 4.880%, 30y 31.4m @ 5.640%	5.81	102.0	5.12%
Ladder	1y 10.0m @ 4.540%, 2y 10.0m @ 4.880%, 3y 10.0m @ 5.000%, 4y 10.0m @ 5.053%, 5y 10.0m @ 5.090%, 6y 10.0m @ 5.139%, 7y 10.0m @ 5.190%, 8y 10.0m @ 5.229%, 9y 10.0m @ 5.260%, 10y 10.0m @ 5.290%	4.57	30.7	5.07%

Table 4. Portfolios priced off the 30 September 2026 curve. All bonds are par bonds (coupon = par yield), so each portfolio is worth $100m today.

The barbell holds 68.6% in the 2-year and 31.4% in the 30-year so that its modified duration equals the bullet's (5.81). It has 2.5 times the bullet's convexity.

7. Six instant scenarios

Scenario	Definition
Parallel +100bp	+100 bp at all maturities
Parallel -100bp	–100 bp at all maturities
Bear steepener	2y unchanged, 30y +100 bp, linear in between; flat outside
Bull flattener	2y unchanged, 30y –100 bp, linear in between; flat outside
Twist	2y +50 bp, 30y –50 bp, linear in between; flat outside
Butterfly	5y–10y +50 bp; 2y and 30y unchanged; linear between 2–5y and 10–30y

Table 5. Scenarios, applied as shifts to continuously compounded zero rates.

Scenario	Bullet: full / dur / dur+cvx	Barbell: full / dur / dur+cvx	Ladder: full / dur / dur+cvx	Barbell – bullet
Parallel +100bp	-5.76 / -5.81 / -5.61	-5.42 / -5.81 / -5.30	-4.54 / -4.57 / -4.42	+34 bp
Parallel -100bp	+6.15 / +5.81 / +6.01	+6.46 / +5.81 / +6.32	+4.84 / +4.57 / +4.73	+30 bp
Bear steepener	-0.97 / -1.04 / -1.03	-2.90 / -4.51 / -4.02	-0.73 / -0.79 / -0.79	-193 bp
Bull flattener	+0.98 / +1.04 / +1.04	+3.66 / +4.51 / +5.01	+0.75 / +0.79 / +0.80	+268 bp
Twist	-1.98 / -1.87 / -1.85	+0.39 / +1.61 / +1.74	-1.59 / -1.49 / -1.48	+237 bp
Butterfly	-2.78 / -2.90 / -2.85	-0.80 / +0.00 / +0.00	-1.91 / -1.99 / -1.96	+198 bp

Table 6. P&L in $m: full revaluation / duration-only / duration + convexity approximation. The approximations use each bond's modified duration and convexity and the scenario shift at the bond's maturity.

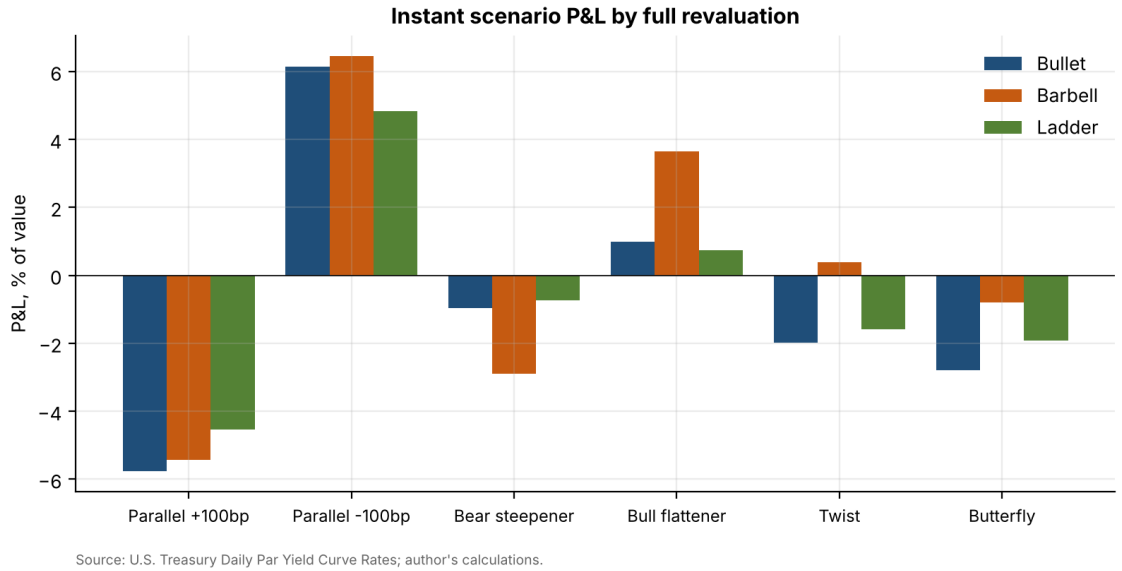


Figure 4. Full-revaluation P&L by scenario.

Parallel moves. With equal durations, the barbell beats the bullet by about 34 bp whether rates rise or fall: that is its convexity. Duration alone misses this; adding convexity recovers most of it.

Curve reshaping dominates. In the bull flattener the barbell's 30-year bond rallies and it gains 268 bp more than the bullet; in the bear steepener it loses 193 bp more. The twist and the butterfly also favour the barbell, because its risk sits in the wings while the bullet's sits in the belly.

When the approximations fail. For non-parallel shifts, applying the shift at each bond's maturity to its yield is crude for a long bond whose early coupons sit at other maturities. In the bear steepener the approximation overstates the barbell's loss (-4.51 vs. -2.90 $m); in the butterfly it shows zero for the barbell, because the shift is zero at 2 and 30 years, while full revaluation shows -0.80 $m, because the 30-year bond's coupons in the 5–10-year sector are hit. Key rate durations fix this.

Portfolio	KRD 2y	KRD 5y	KRD 10y	KRD 30y	Sum of KRDs	Modified duration
Bullet	0.30	3.58	2.07	-0.00	5.95	5.81
Barbell	1.42	0.29	1.35	2.83	5.90	5.81
Ladder	0.80	2.07	1.81	-0.00	4.69	4.57

Table 7. Key rate durations (triangular 1 bp bumps of the zero curve at 2, 5, 10 and 30 years). They sum to about (1 + y/2) times modified duration because they are defined on continuous zero rates.

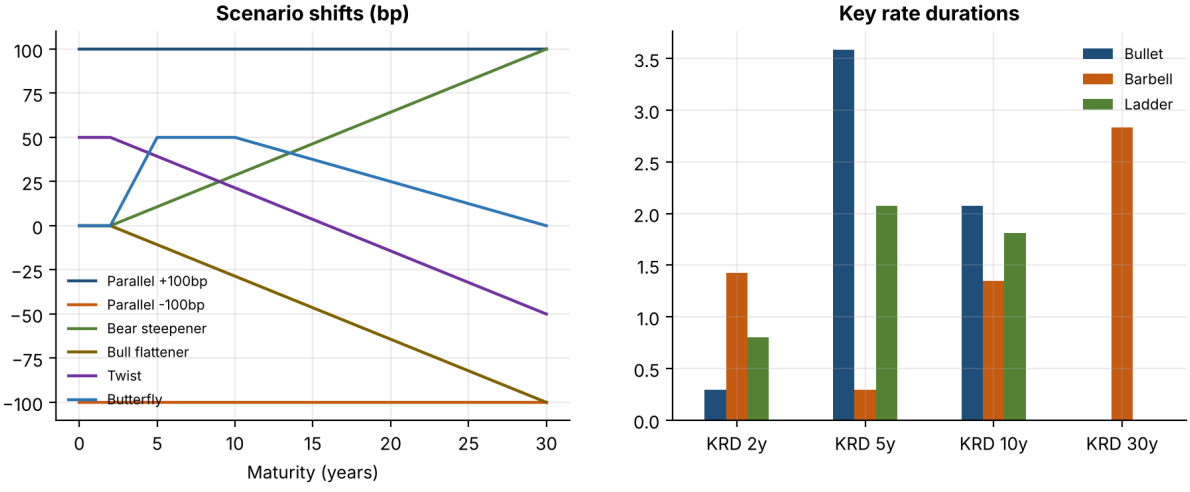


Figure 5. Scenario shifts and key rate durations.

8. Twelve months of carry and roll-down

Portfolio	Total	Carry (yield unchanged)	Roll-down
Bullet	5.45%	5.19%	0.26%
Barbell	5.27%	5.12%	0.15%
Ladder	5.26%	5.07%	0.20%

Table 8. 12-month return if the curve does not move. Carry holds each bond's yield constant; roll-down is the extra gain from bonds rolling down an upward-sloping curve. Coupons are not reinvested; the ladder's 1-year bond matures.

The bullet earns the most: 5.45%, against 5.27% for the barbell. The barbell gives up 19 bp a year: a lower coupon on the 2-year (4.88%) and negative roll-down on the 30-year, which sits on the inverted 20–30-year segment and loses value as it rolls towards the higher-yielding 20-year point. Convexity is not free.

Conclusion

A bootstrapped curve reprices the market exactly, while an NSS fit gives a smooth curve close to the Fed's. Three factors drive almost all yield changes, and for a bond portfolio the key choice is not duration but where on the curve it sits. A duration-matched barbell is a bet on flattening and on volatility: it gained 268 bp over the bullet in a bull flattener and lost 193 bp in a bear steepener, at a cost of 19 bp of carry a year if nothing happens.

Limitations

- Scenarios are instantaneous and ignore the carry earned while curves move.
- Par yields are indicative closing levels for on-the-run securities; no bid-ask, repo or liquidity premia.
- The daily Nelson-Siegel fit uses a fixed decay and par yields rather than zero rates, which is standard but approximate.
- Shifts are applied to zero rates; applying them to par yields would give slightly different numbers.

References

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